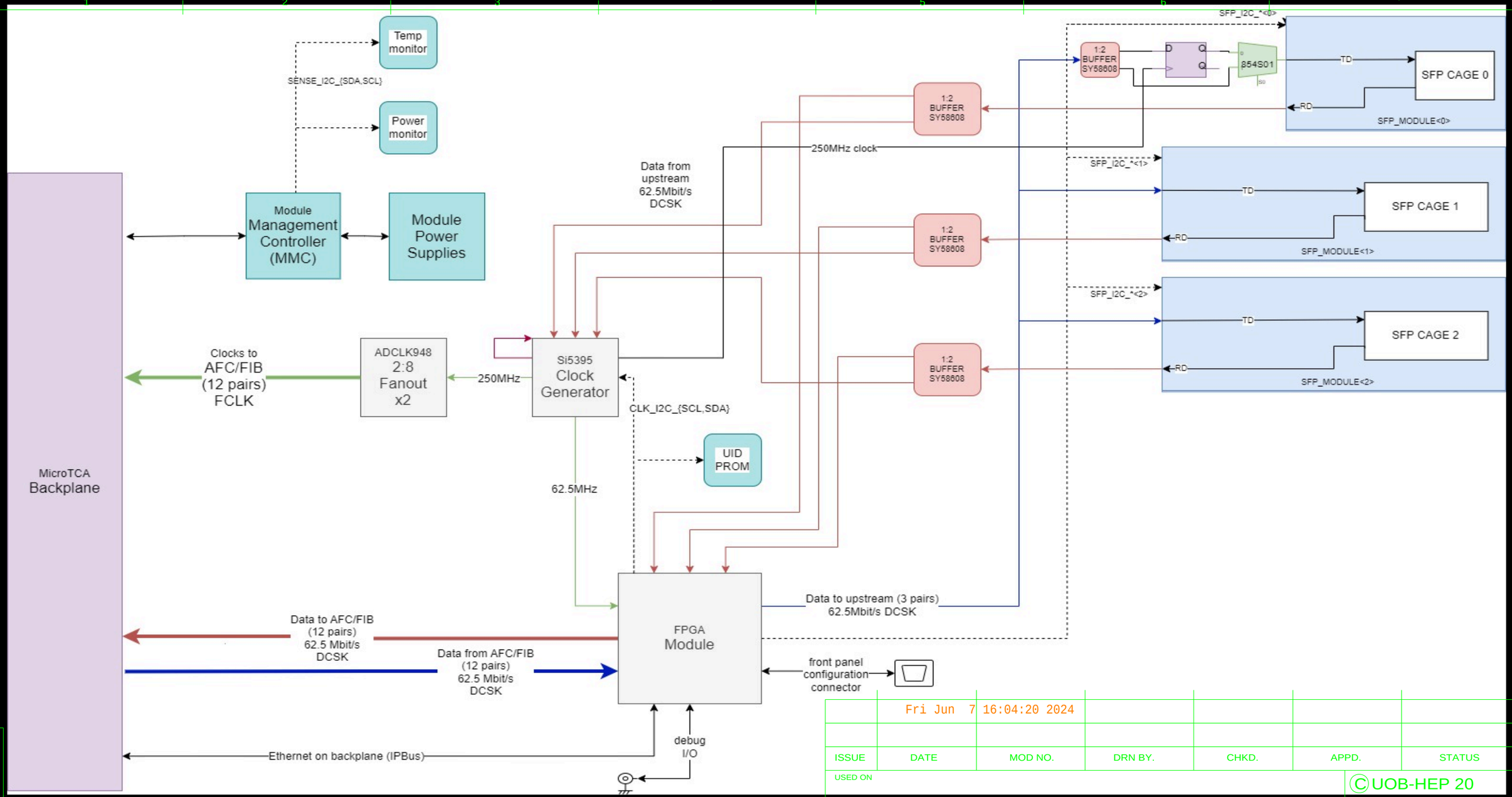


N.B. THIS IS A DRAWING, NO CONNECTIVITY

	Fri Jun 7 16:04:20 2024					
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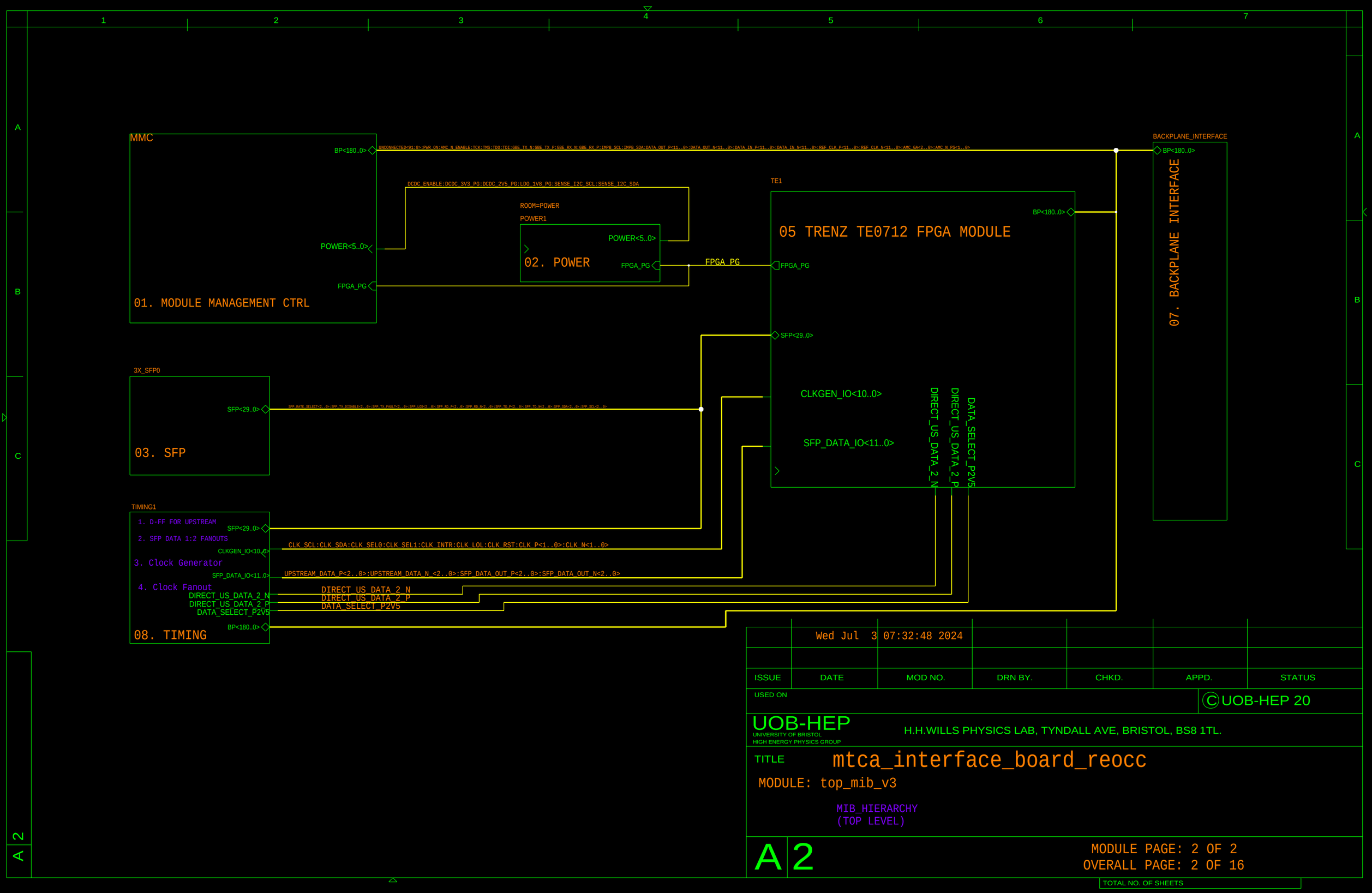
TITLE mtca_interface_board_reocc

MODULE: top_mib_v3

A2

MODULE PAGE: 1 OF 2
OVERALL PAGE: 1 OF 16

TOTAL NO. OF SHEETS



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TITLE mtca_interface_board_reocc

MODULE: top_mib_v3

MIB_HIERARCHY
(TOP LEVEL)

A2

MODULE PAGE: 2 OF 2
OVERALL PAGE: 2 OF 16

TOTAL NO. OF SHEETS

12V TO 3V3 @ 14A DC-DC CONVERTER

MIN ESR OF 3 MILLI-OHMS ON OUTPUT OF PTH08T210WAD

WANT < 0.3V DROP AT 2A

VP3V3 IV MONITOR

12V @ 6.6A IN

3.3V @ 14A OUT

VP12

C25 330UF

C51 330UF

C42 0.1UF 25V

I193

I195

POWER1

PTH08T210WAD

TRACK

TT

+SENS

VI[2]

VO[2]

VO_3V3<1..0>

UVLO

-SENS

GND[4]

VADJ

ECP

VIN<1..0>

DC_DC_ENABLE

R33 150K

R79 XX 1%

R84 1 1%

C75 0.1UF 25V

C92 0.1UF 25V

C95 100UF 6.3V

C17 220UF

C44 220UF

VCC3V3_PRE_RSHUNT

TP5

BOM_IGNORE=TRUE

R62 0.05

C61 220UF

C63 220UF

VCC3V3

IC5 TLP183 SOIC

T1 PJE8408 SOT523

R70 1.6K 1%

R16 24.9

R31 24.9

MJB1

MP3V3

Default: 2-3

J12

VP12_EN

DCDC_ENABLE

HDR1x3

DCDC FORCE ENABLE

SK1

EXT POWER IN KK CONNECTOR

SENSE_12C_SDA

SENSE_12C_SCL

VCC=MP3V3

IC6 LTC2991CMS#PBF

SDA

SCL

ADR0

ADR1

ADR2

PWM

V1

V2

V3

V4

V5

V6

V7

V8

MSOP

R82

R64

R66

P2V5

VP1V8

1V8_POST_RSHUNT

P2V5_MON_HIGH

P2V5_MON_LOW

MEASURE CURRENT THROUGH
SENSE RESISTOR BEFORE 1V8 REG
MEASURE 1V8 VOLTAGE AFTER REGULATOR

I2C ADDRESS = 90H

C91 0.1UF 25V

DECOUPLE
LTC2991

R135 249 Ohm

R85 10K

R86 4.99k

D4 Red

DCDC_3V3_PG

C69 0.1UF 25V

C71 47UF 6.3V

C81 47UF 6.3V

C82 47UF 6.3V

C50 47UF 6.3V

TP7

TP13

ISSUE	DATE	MOD NO.	DRN BY.	CHKD.	APPD.	STATUS
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TITLE mtca_interface_board_reocc

MODULE: 02_ power

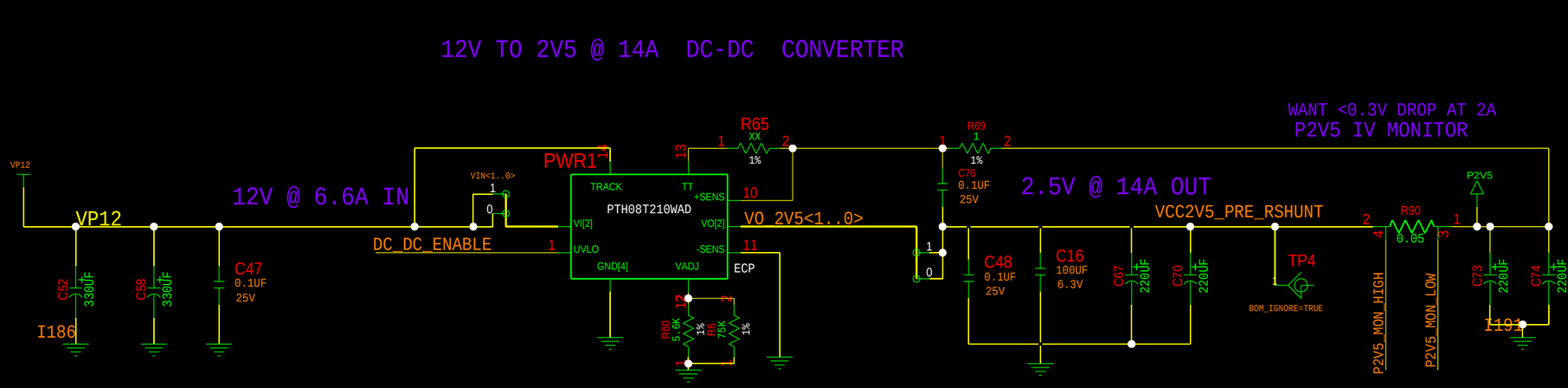
DC-DC (12V --> 3.3V)

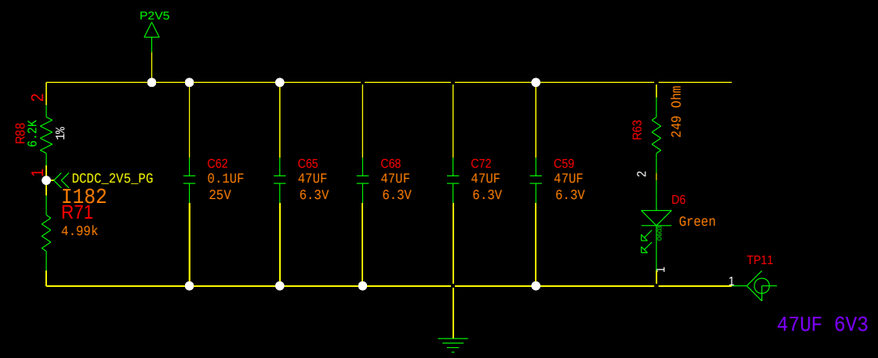
A2

MODULE PAGE: 1 OF 3
OVERALL PAGE: 3 OF 16

TOTAL NO. OF SHEETS

12V TO 2V5 @ 14A DC-DC CONVERTER





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TITLE mtca_interface_board_reocc

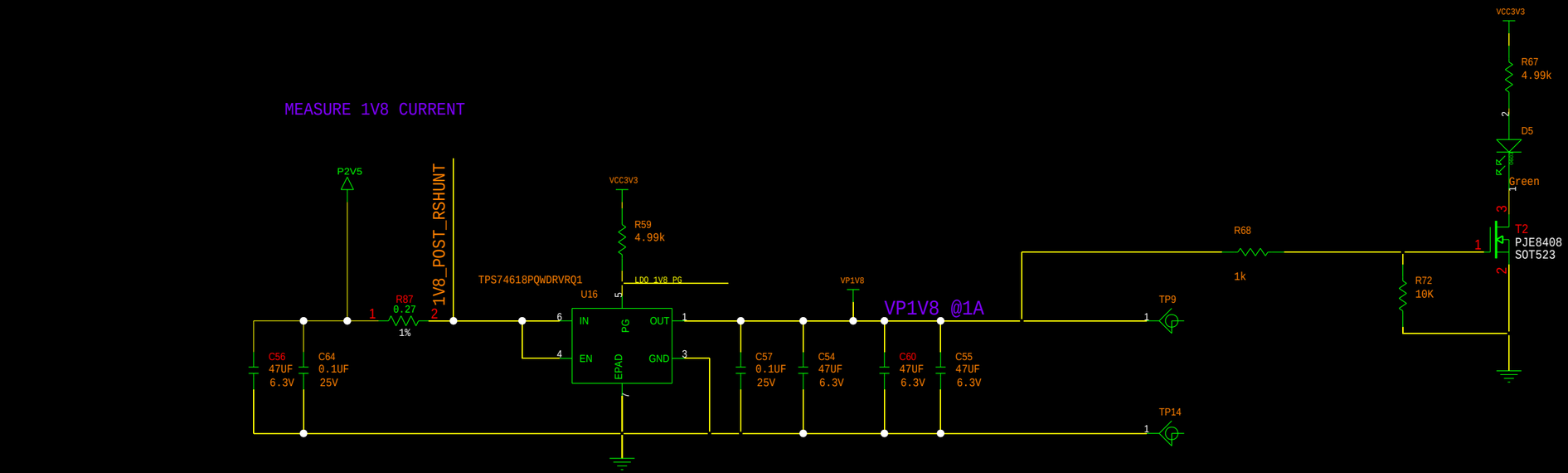
MODULE: 02_ power

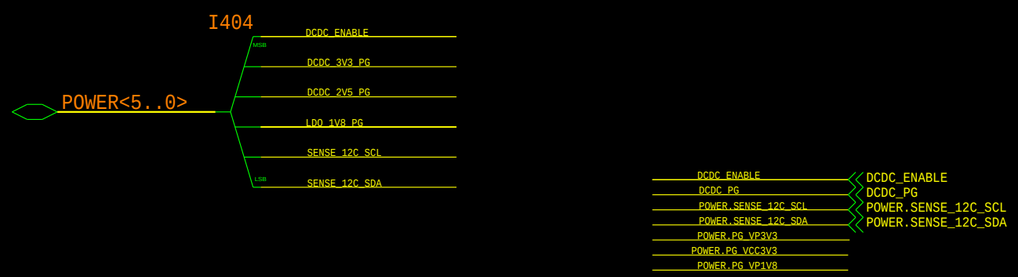
DC-DC (12V --> 2.5V)

A2

MODULE PAGE: 2 OF 3
OVERALL PAGE: 4 OF 16

TOTAL NO. OF SHEETS





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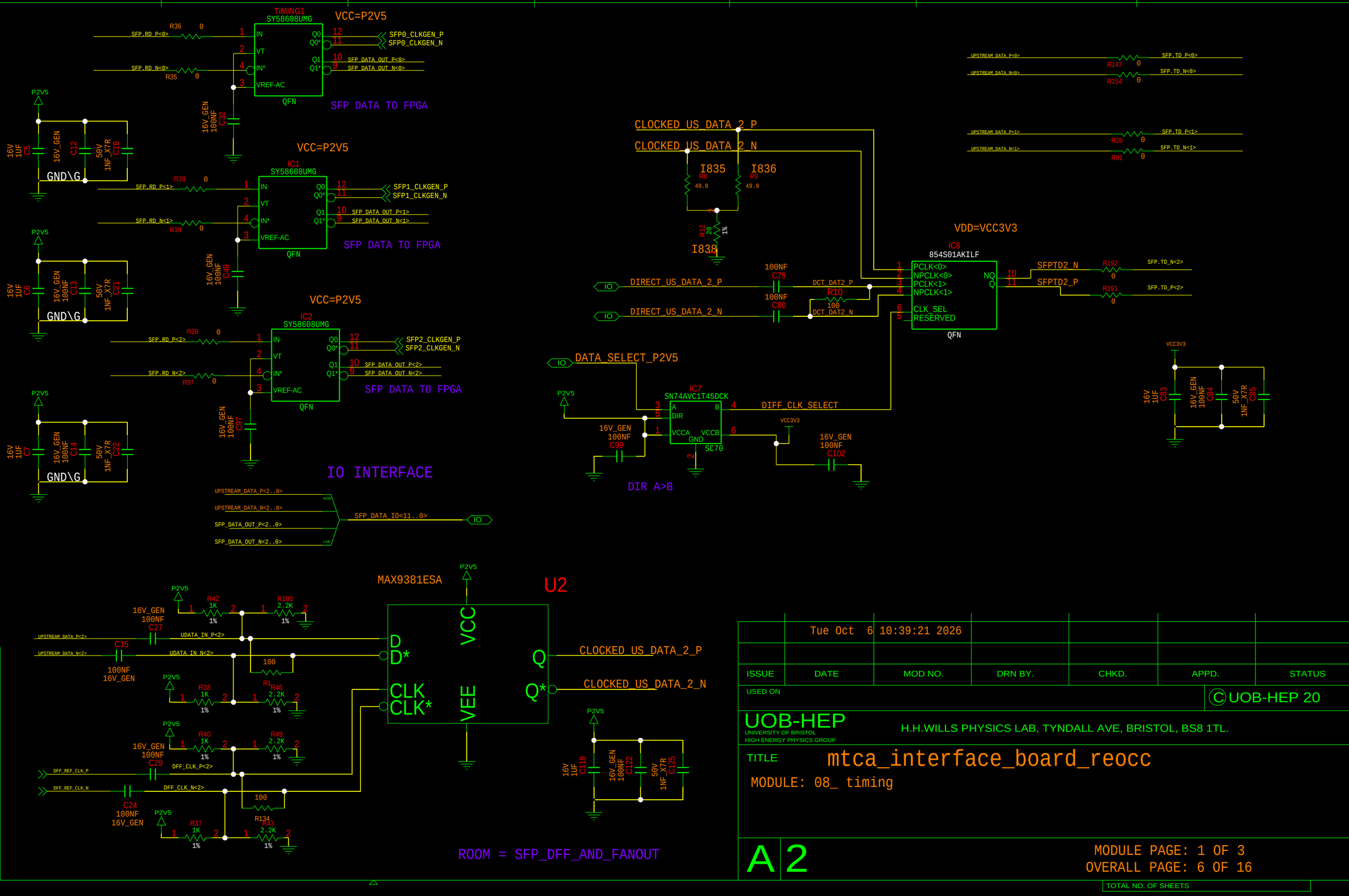
TITLE mtca_interface_board_reocc

MODULE: 02_ power

A 2

MODULE PAGE: 3 OF 3
OVERALL PAGE: 5 OF 16

TOTAL NO. OF SHEETS



	Tue Oct 6 10:39:21 2026					
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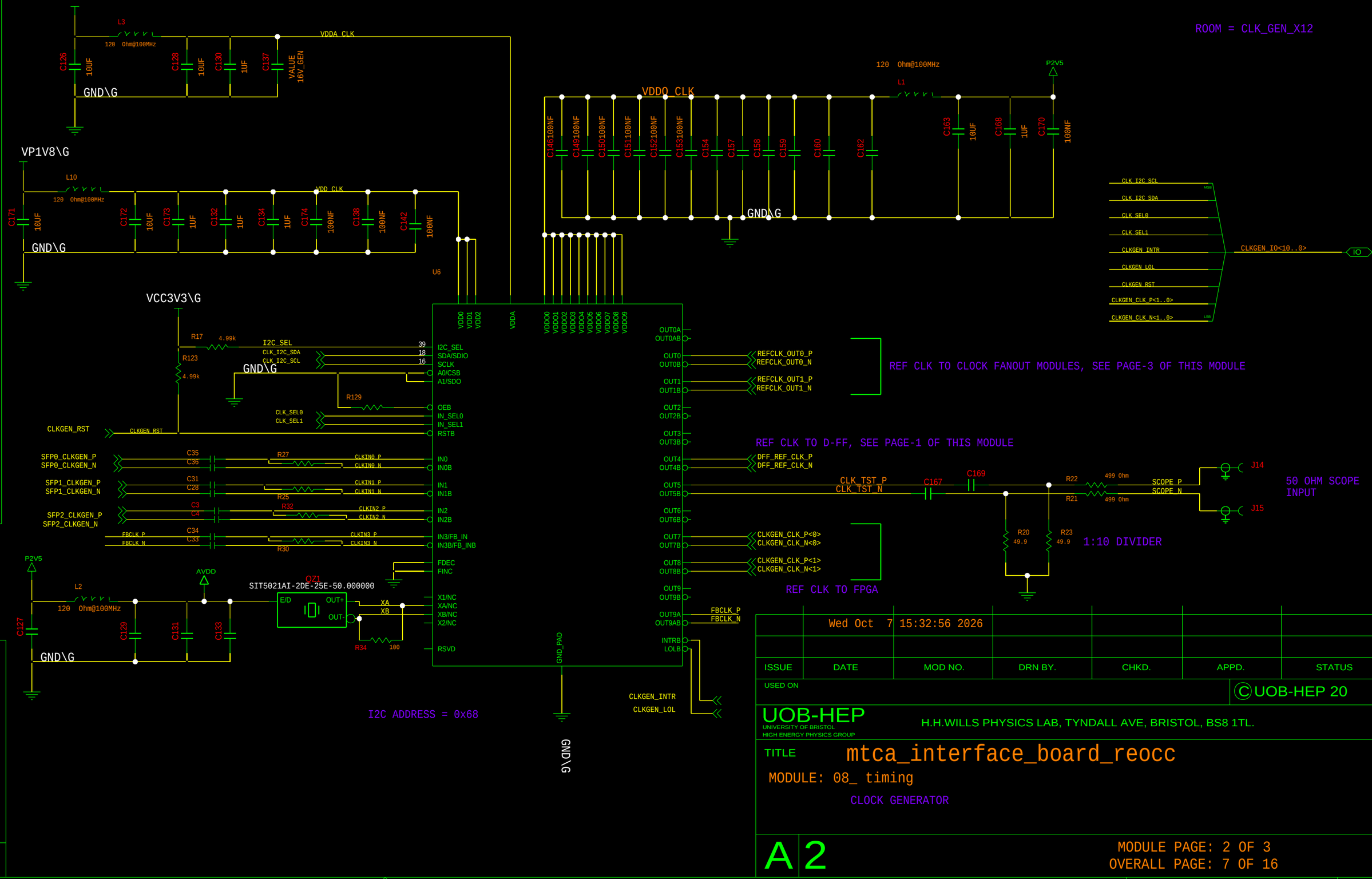
TITLE **mtca_interface_board_reocc**

MODULE: 08_ timing

A 2

MODULE PAGE: 1 OF 3
OVERALL PAGE: 6 OF 16

TOTAL NO. OF SHEETS



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TITLE mtca_interface_board_reocc

MODULE: 08_ timing

CLOCK GENERATOR

A2

VEE=GND;VCC=VDD_CLK_BUFFER

IC3 ADCLK948BCPZ

I180

C187 100NF 16V_GEN

C10 0.1UF 25V

C11 0.1UF 25V

REFCLK_OUT0_P
REFCLK_OUT0_N

RCLK_P0
RCLK_N0

VREF0 VT0 CLK0 CLK0* VT1 CLK1 CLK1* IN_SEL VREF1

LFCSP

R11 1K 1%

I213

VDD_CLK_BUFFER\G

R155 1K 1%

R157 2.2K 1%

Q[0] Q[0]* Q[1] Q[1]* Q[2] Q[2]* Q[3] Q[3]* Q[4] Q[4]* Q[5] Q[5]* Q[6] Q[6]* Q[7] Q[7]*

DC_REF_CLK_P<0>
DC_REF_CLK_N<0>
DC_REF_CLK_P<1>
DC_REF_CLK_N<1>
DC_REF_CLK_P<2>
DC_REF_CLK_N<2>
DC_REF_CLK_P<3>
DC_REF_CLK_N<3>
DC_REF_CLK_P<4>
DC_REF_CLK_N<4>
DC_REF_CLK_P<5>
DC_REF_CLK_N<5>

R161 R165 R169 R173 R177 R181 R185 R189
R159 R163 R167 R171 R175 R179 R183 R187

200 1% 2 1

C188 C189 C190 C191 C192 C193 C194 C195 C196 C197 C198 C199 100NF 16V_GEN

BP.REF_CLK_P<0>
BP.REF_CLK_N<0>
BP.REF_CLK_P<1>
BP.REF_CLK_N<1>
BP.REF_CLK_P<2>
BP.REF_CLK_N<2>
BP.REF_CLK_P<3>
BP.REF_CLK_N<3>
BP.REF_CLK_P<4>
BP.REF_CLK_N<4>
BP.REF_CLK_P<5>
BP.REF_CLK_N<5>

ROOM = CLK_FO_0_5

VEE=GND;VCC=VDD_CLK_BUFFER

IC4 ADCLK948BCPZ

I247

C139 100NF 16V_GEN

C26 0.1UF 25V

C30 0.1UF 25V

REFCLK_OUT1_P
REFCLK_OUT1_N

RCLK_P1
RCLK_N1

VREF0 VT0 CLK0 CLK0* VT1 CLK1 CLK1* IN_SEL VREF1

LFCSP

R24 1K 1%

I254

VDD_CLK_BUFFER\G

R156 1K 1%

R158 2.2K 1%

Q[0] Q[0]* Q[1] Q[1]* Q[2] Q[2]* Q[3] Q[3]* Q[4] Q[4]* Q[5] Q[5]* Q[6] Q[6]* Q[7] Q[7]*

DC_REF_CLK_P<6>
DC_REF_CLK_N<6>
DC_REF_CLK_P<7>
DC_REF_CLK_N<7>
DC_REF_CLK_P<8>
DC_REF_CLK_N<8>
DC_REF_CLK_P<9>
DC_REF_CLK_N<9>
DC_REF_CLK_P<10>
DC_REF_CLK_N<10>
DC_REF_CLK_P<11>
DC_REF_CLK_N<11>

R162 R166 R170 R174 R178 R182 R186 R190
R160 R164 R168 R172 R176 R180 R184 R188

200 1% 2 1

I243

C175 C176 C177 C178 C179 C180 C181 C182 C183 C184 C185 C186 100NF 16V_GEN

BP.REF_CLK_P<6>
BP.REF_CLK_N<6>
BP.REF_CLK_P<7>
BP.REF_CLK_N<7>
BP.REF_CLK_P<8>
BP.REF_CLK_N<8>
BP.REF_CLK_P<9>
BP.REF_CLK_N<9>
BP.REF_CLK_P<10>
BP.REF_CLK_N<10>
BP.REF_CLK_P<11>
BP.REF_CLK_N<11>

ROOM = CLK_FO_6_11

I214 VCC3V3

L9 BLM21PG221SN1

VDD_CLK_BUFFER\G

GND\G

C112 10UF 10V

C121 10UF 10V C37 100NF 16V_GEN

C38 1UF 6.3V C135 100NF 16V_GEN C19 100NF 16V_GEN

C20 1UF 6.3V C136 100NF 16V_GEN

C155 10UF 10V C148 100NF 16V_GEN

C165 1UF 6.3V C166 100NF 16V_GEN C167 100NF 16V_GEN

C9 1UF 6.3V C8 100NF 16V_GEN

I177 SFP<29..0>

SFP<29..0>

SFP_RATE_SELECT<2..0>
SFP_TX_DISABLE<2..0>
SFP_TX_FAULT<2..0>
SFP_LOS<2..0>
SFP_RD_P<2..0>
SFP_RD_N<2..0>
SFP_TD_P<2..0>
SFP_TD_N<2..0>
SFP_SDA<2..0>
SFP_SCL<2..0>

BP<180..0>

BP.REF_CLK_P<11..0>
BP.REF_CLK_N<11..0>

USE 8 WAY LVPECL FANOUT
... BECAUSE 6 WAY LVDS BUFFER
NOT AVAILABLE DUE TO
SUPPLY CHAIN ISSUES

ADCLK984 SUPPLY CURRENT < 330MA
10MA PER OUTPUT PIN = 160MA
TOTAL PER DEVICE = 490MA
TOTAL FOR FANOUT = 980MA

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TITLE mtca_interface_board_reocc

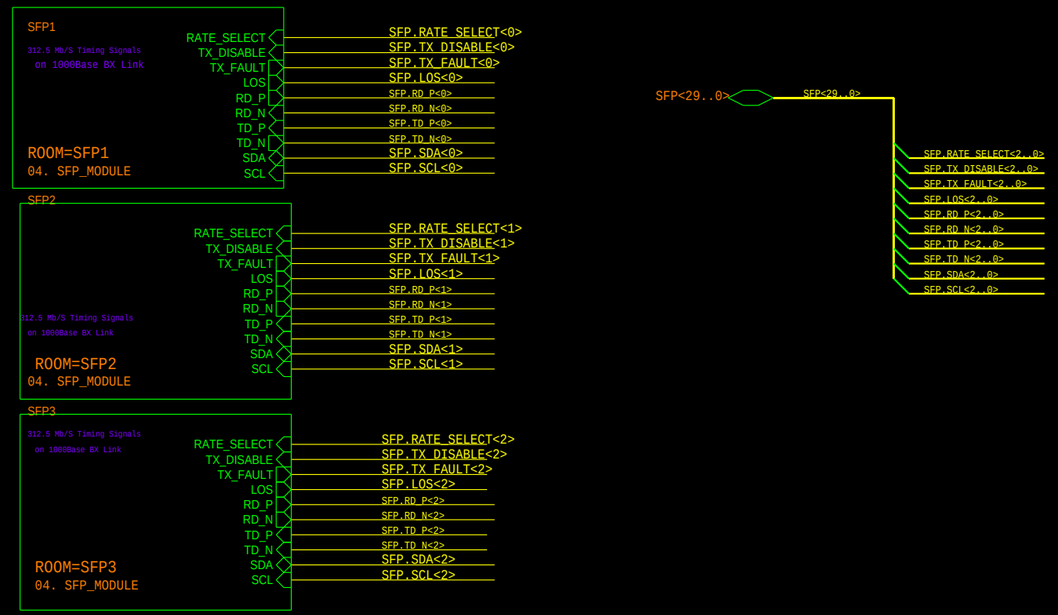
MODULE: 08_ timing

CLOCK FANOUT

A2

MODULE PAGE: 3 OF 3
OVERALL PAGE: 8 OF 16

TOTAL NO. OF SHEETS



	Mon Mar 6 16:31:19 2023					
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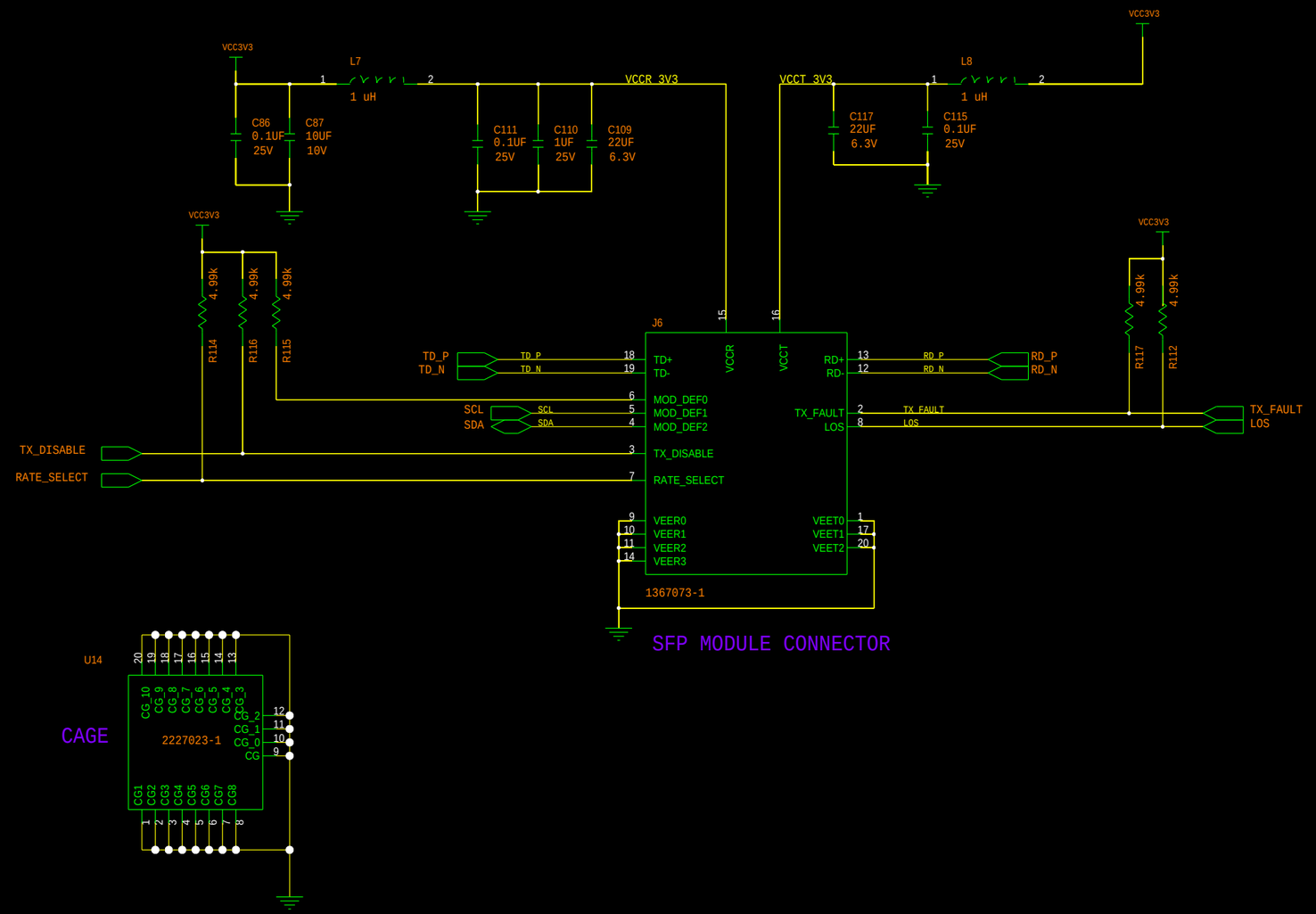
TITLE mtca_interface_board_reocc

MODULE: 03_ sfp

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 9 OF 16

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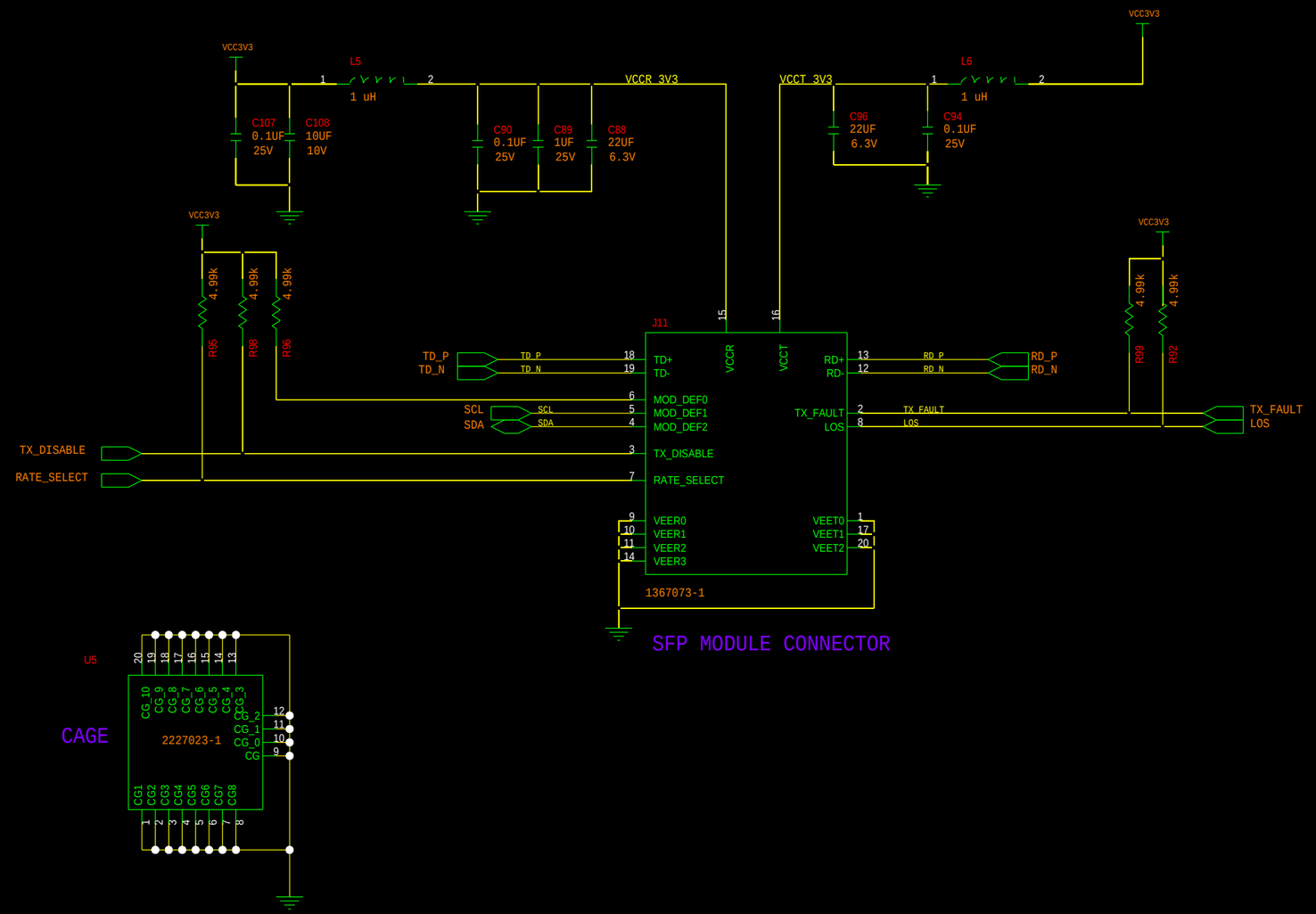
TITLE mtca_interface_board_reocc

MODULE: 04_ sfp_module

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 10 OF 16

TOTAL NO. OF SHEETS



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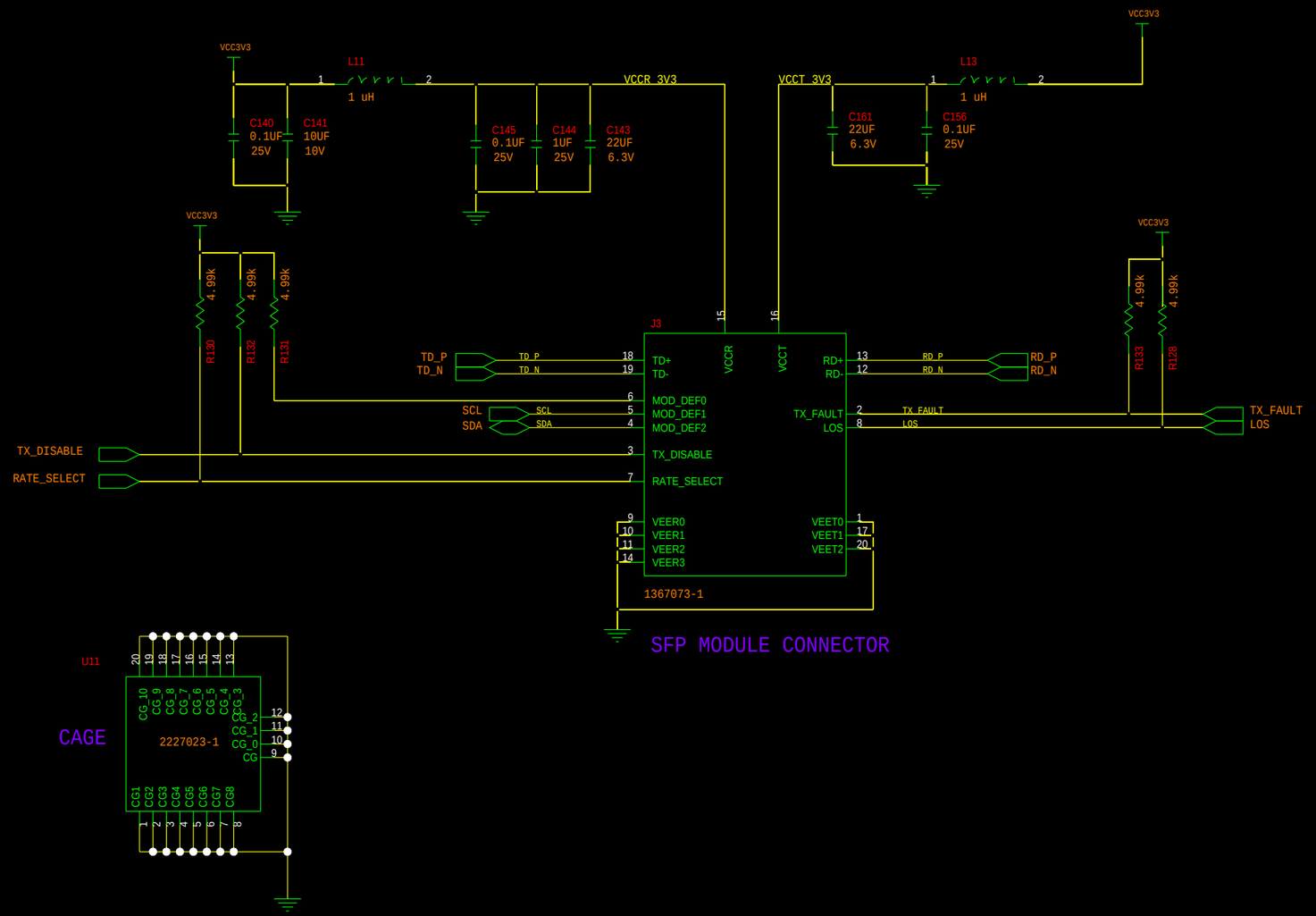
TITLE mtca_interface_board_reocc

MODULE: 04_ sfp_module

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 11 OF 16

TOTAL NO. OF SHEETS



SFP MODULE CONNECTOR

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TITLE mtca_interface_board_reocc

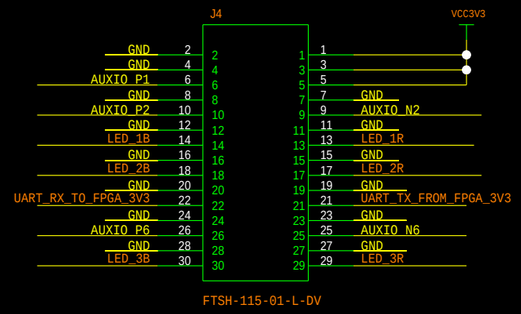
MODULE: 04_ sfp_module

A2

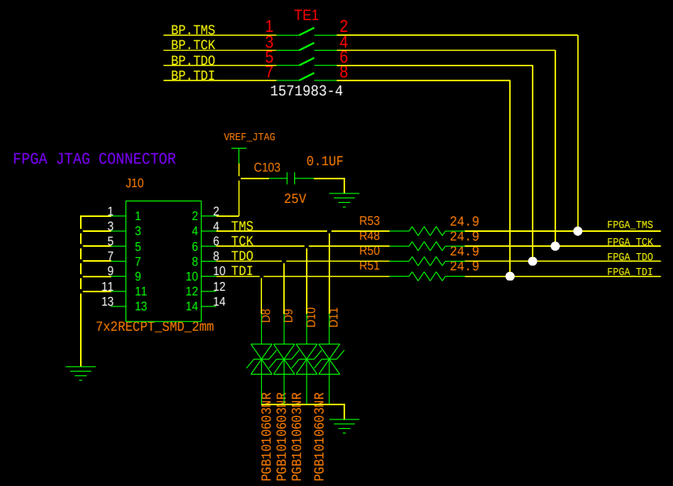
MODULE PAGE: 1 OF 1
OVERALL PAGE: 12 OF 16

TOTAL NO. OF SHEETS

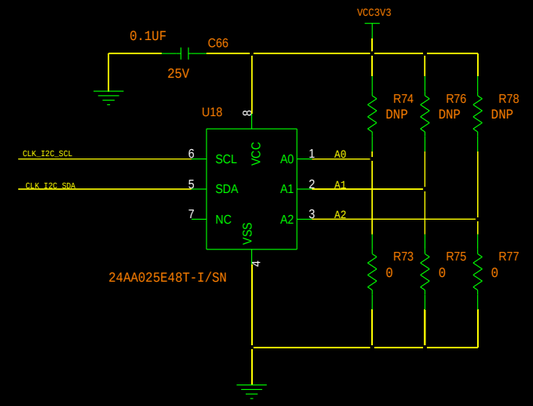
AUX IO INTERFACE



FPGA JTAG CONNECTOR

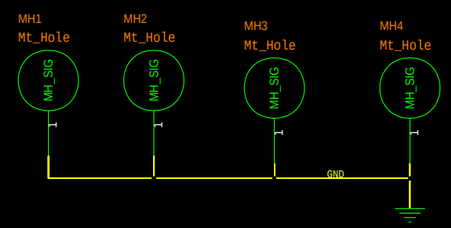


EEPROM

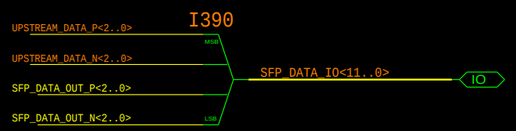


I2C ADDRESS = 0X50

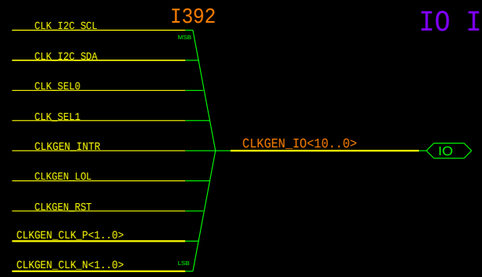
TRENZ TE0712 FPGA MODULE MOUNT HOLES



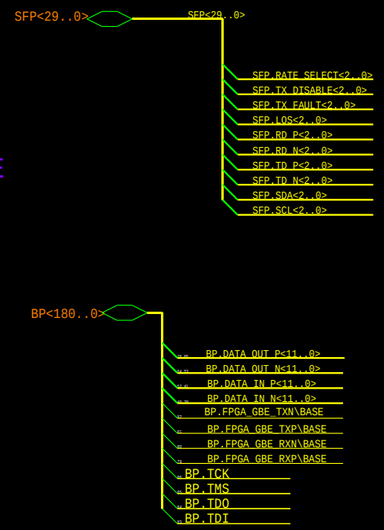
I390



I392



IO INTERFACE



ISSUE	DATE	MOD NO.	DRN BY.	CHKD.	APPD.	STATUS
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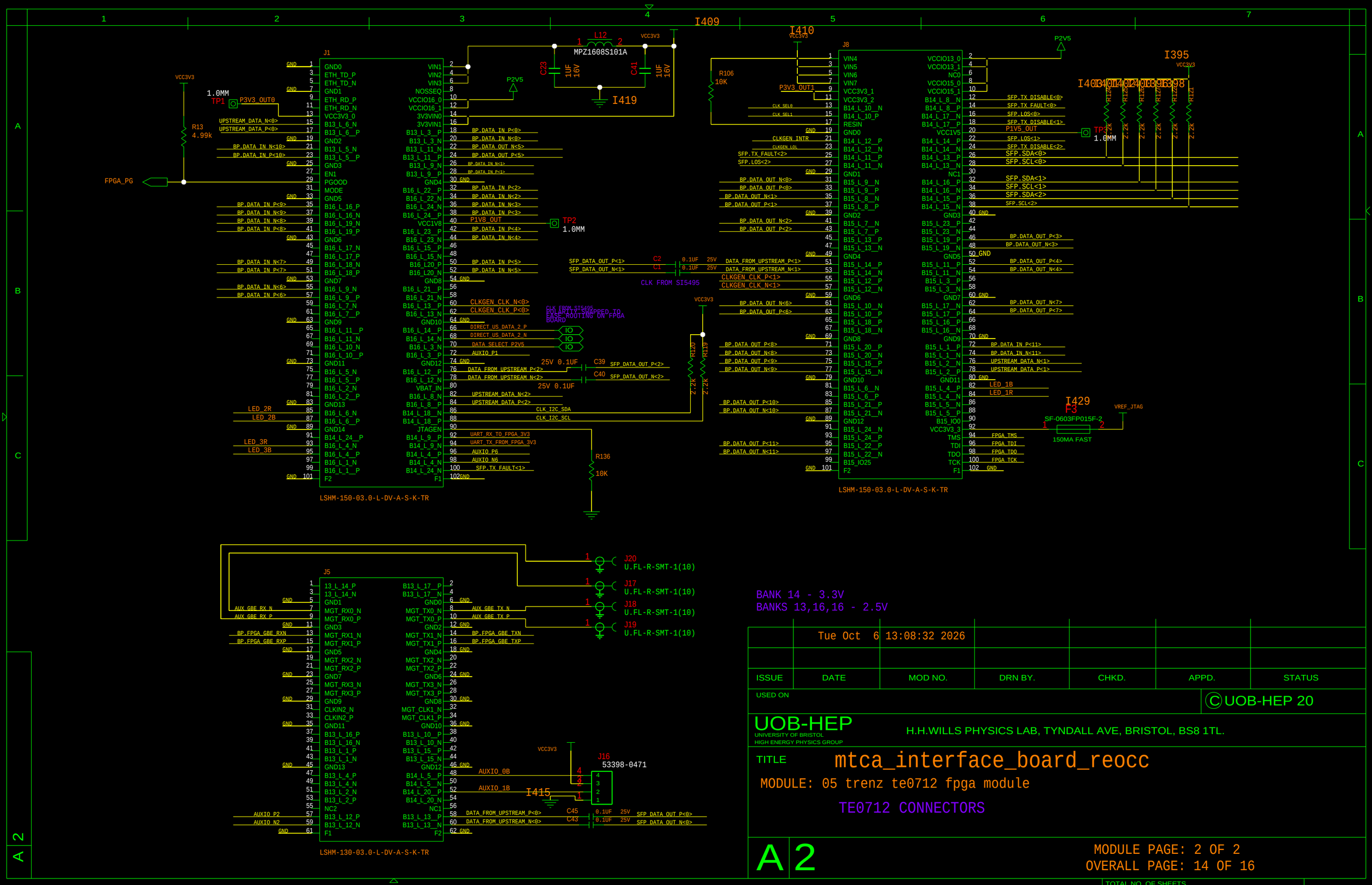
TITLE **mtca_interface_board_reocc**

MODULE: 05 trenz te0712 fpga module

A2

MODULE PAGE: 1 OF 2
OVERALL PAGE: 13 OF 16

TOTAL NO. OF SHEETS



	Tue Oct 6 13:08:32 2026					
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TITLE **mtca_interface_board_reocc**

MODULE: 05 trenz te0712 fpga module

TE0712 CONNECTORS

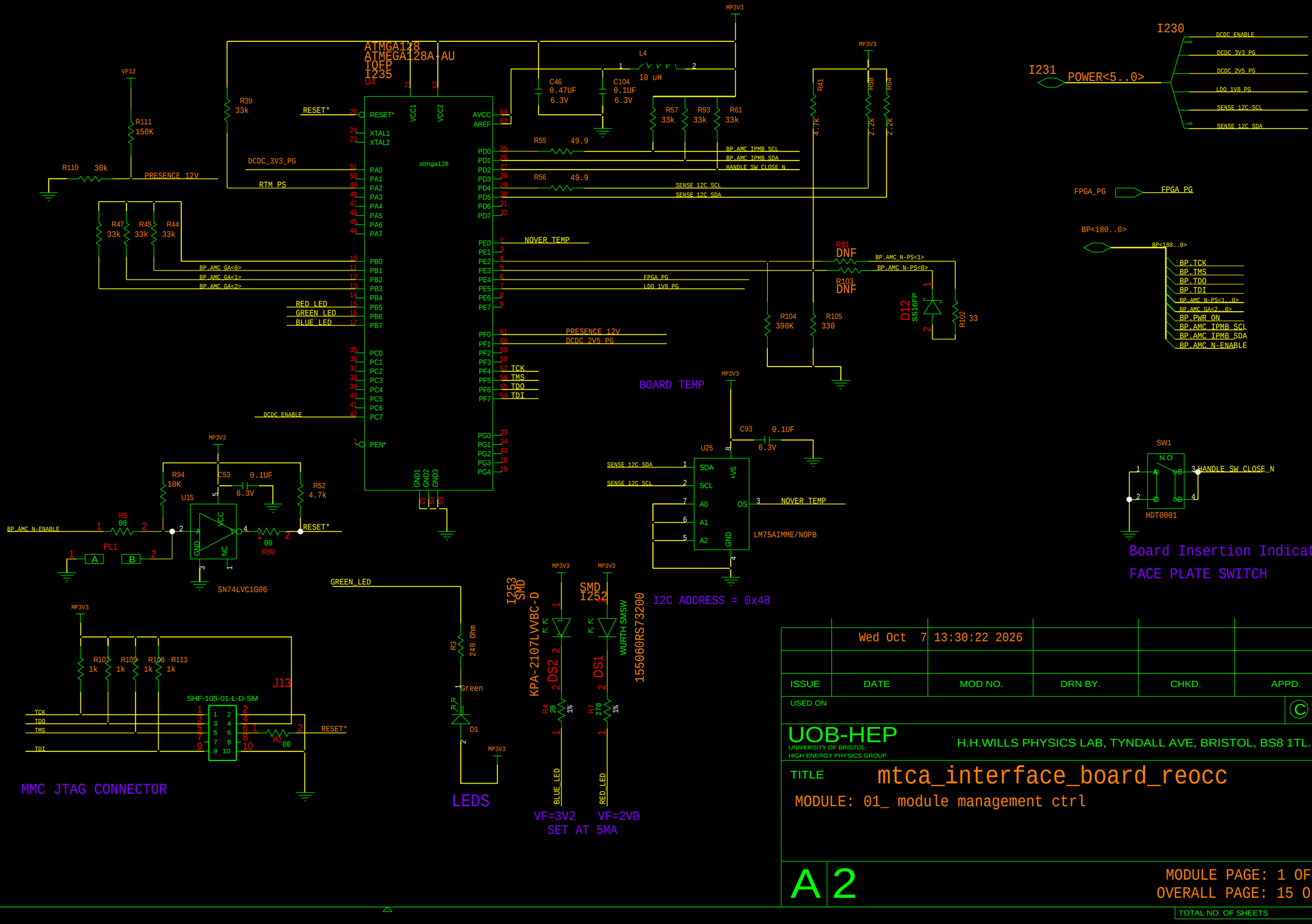
A 2

MODULE PAGE: 2 OF 2
OVERALL PAGE: 14 OF 16

TOTAL NO. OF SHEETS

MODULE MANAGEMENT CONTROL

IO INTERFACE



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TITLE **mtca_interface_board_reocc**

MODULE: 01_ module management ctrl

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 15 OF 16

TOTAL NO. OF SHEETS

TONGUE 2 MEZZENINE CONNECTOR

VP12

F1
SF-0603S300-2
3A

MP3V3

F2
SF-0603FP015F-2
150MA FAST

J7

GND	1	GND1	PWR_ON	170	BP.PWR_ON
12V_IN	2	PWR1	TDI	169	BP.TDI
BP.AMC_N-PS<1>	3	PS1#	TDO	168	BP.TDO
MP3V3_IN	4	MP	TRST#	167	
BP.AMC_GA<0>	5	GA0	TMS	166	BP.TMS
	6	RSVD0	TCK	165	BP.TCK
GND	7	GND2	GND55	164	GND
	8	RSVD1	TXFA-1+	163	
12V_IN	9	PWR2	TXFA-1-	162	
GND	10	GND3	GND54	161	GND
BP.FPGA_GBE_TXP	11	TXFUA+	RXFA-1+	160	
BP.FPGA_GBE_TXN	12	TXFUA-	RXFA-1-	159	
GND	13	GND4	GND53	158	GND
BP.FPGA_GBE_RXP	14	RXFUA+	TXFA-2+	157	
BP.FPGA_GBE_RXN	15	RXFUA-	TXFA-2-	156	
GND	16	GND5	GND52	155	GND
BP.AMC_GA<1>	17	GA1	RXFA-2+	154	
12V_IN	18	PWR3	RXFA-2-	153	
GND	19	GND6	GND51	152	GND
	20	TXFA-3+	TXFA-4+	151	
	21	TXFA-3-	TXFA-4-	150	
GND	22	GND7	GND50	149	GND
	23	RXFA-3+	RXFA-4+	148	
	24	RXFA-3-	RXFA-4-	147	
GND	25	GND8	GND49	146	GND
BP.AMC_GA<2>	26	GA2	TXFA-6+	145	
12V_IN	27	PWR4	TXFA-6-	144	
GND	28	GND9	GND48	143	GND
	29	TXFA-5+	RXFA-6+	142	
	30	TXFA-5-	RXFA-6-	141	
GND	31	GND10	GND47	140	GND
	32	RXFA-5+	TXFA-8+	139	
	33	RXFA-5-	TXFA-8-	138	
GND	34	GND11	GND46	137	GND
	35	TXFA-7+	RXFA-8+	136	
	36	TXFA-7-	RXFA-8-	135	
GND	37	GND12	GND45	134	GND
	38	RXFA-7+	TMREQ#	133	
	39	RXFA-7-	RSVD	132	
GND	40	GND13	GND44	131	GND
BP.AMC_N-ENABLE	41	ENABLE#	I2C_SCL	130	
12V_IN	42	PWR5	I2C_SDA	129	
GND	43	GND14	GND43	128	GND
	44	TXFA-9+	IPMB0-SCL-A	127	
	45	TXFA-9-	IPMB0-SDA-A	126	
GND	46	GND15	GND42	125	GND
	47	RXFA-9+	IPMB0-SCL-B	124	
	48	RXFA-9-	IPMB0-SDA-B	123	
GND	49	GND16	GND41	122	GND
	50	TXFA-10+	IPMBL-SCL-1	121	
	51	TXFA-10-	IPMBL-SDA-1	120	
GND	52	GND17	GND40	119	GND
	53	RXFA-10+	IPMBL-SCL-2	118	
	54	RXFA-10-	IPMBL-SDA-2	117	
GND	55	GND18	GND39	116	GND
BP.AMC_IPMB_SCL	56	SCL_L	IPMBL-SCL-3	115	
12V_IN	57	PWR6	IPMBL-SDA-3	114	
GND	58	GND19	GND38	113	GND
	59	TXFA-11+	IPMBL-SCL-4	112	
	60	TXFA-11-	IPMBL-SDA-4	111	
GND	61	GND20	GND37	110	GND
	62	RXFA-11+	IPMBL-SCL-5	109	
	63	RXFA-11-	IPMBL-SDA-5	108	
GND	64	GND21	GND36	107	GND
	65	TXFA-12+	IPMBL-SCL-6	106	
	66	TXFA-12-	IPMBL-SDA-6	105	
GND	67	GND22	GND35	104	GND
	68	RXFA-12+	IPMBL-SCL-7	103	
	69	RXFA-12-	IPMBL-SDA-7	102	
GND	70	GND23	GND34	101	GND
BP.AMC_IPMB_SDA	71	SDA_L	IPMBL-SCL-8	100	
12V_IN	72	PWR7	IPMBL-SDA-8	99	
GND	73	GND24	GND33	98	GND
	74	XOVER0+	IPMBL-SCL-9	97	
	75	XOVER0-	IPMBL-SDA-9	96	
GND	76	GND25	GND32	95	GND
	77	XOVER1+	IPMBL-SCL-10	94	
	78	XOVER1-	IPMBL-SDA-10	93	
GND	79	GND26	GND31	92	GND
	80	XOVER2+	IPMBL-SCL-11	91	
	81	XOVER2-	IPMBL-SDA-11	90	
GND	82	GND27	GND30	89	GND
BP.AMC_N-PS<0>	83	PS0#	IPMBL-SCL-12	88	
12V_IN	84	PWR8	IPMBL-SDA-12	87	
GND	85	GND28	GND29	86	GND

MCH_TG1-EDGE-170

BP PLANE CONNECTOR TONGUE 1

R80
0

J2

GND	2	1	GND
BP.REF_CLK_P<0>	4	3	GND
BP.REF_CLK_N<0>	6	5	GND
GND	8	7	BP.REF_CLK_P<1>
GND	10	9	BP.REF_CLK_N<1>
BP.REF_CLK_P<2>	12	11	GND
BP.REF_CLK_N<2>	14	13	GND
GND	16	15	BP.REF_CLK_P<3>
GND	18	17	BP.REF_CLK_N<3>
BP.REF_CLK_P<4>	20	19	GND
BP.REF_CLK_N<4>	22	21	GND
GND	24	23	BP.REF_CLK_P<5>
GND	26	25	BP.REF_CLK_N<5>
BP.REF_CLK_P<6>	28	27	GND
BP.REF_CLK_N<6>	30	29	GND
GND	32	31	BP.REF_CLK_P<7>
GND	34	33	BP.REF_CLK_N<7>
BP.REF_CLK_P<8>	36	35	GND
BP.REF_CLK_N<8>	38	37	GND
GND	40	39	BP.REF_CLK_P<9>
GND	42	41	BP.REF_CLK_N<9>
BP.REF_CLK_P<10>	44	43	GND
BP.REF_CLK_N<10>	46	45	GND
GND	48	47	BP.REF_CLK_P<11>
GND	50	49	BP.REF_CLK_N<11>
GND	52	51	GND
GND	54	53	GND
GND	56	55	GND
GND	58	57	GND
GND	60	59	GND
GND	62	F1	GND
		F2	

LSHM-130-03.0-L-DV-A-S-K-TR

BP<180..0>

IO INTERFACE

BP.PWR_ON
BP.AMC_N-ENABLE
BP.TCK
BP.TMS
BP.TDO
BP.TDI
BP.FPGA_GBE_TXN
BP.FPGA_GBE_TXP
BP.FPGA_GBE_RXN
BP.FPGA_GBE_RXP
BP.AMC_IPMB_SCL
BP.AMC_IPMB_SDA
BP.DATA_OUT_P<11..0>
BP.DATA_OUT_N<11..0>
BP.DATA_IN_P<11..0>
BP.DATA_IN_N<11..0>
BP.REF_CLK_P<11..0>
BP.REF_CLK_N<11..0>
BP.AMC_GA<2..0>
BP.AMC_N-PS<1..0>

ESD PROTECTION

SEGMENT2
SEGMENT1
SEGMENT3

R138
10M
R137
10M

Promotes save ESD discharge
during board insertion (see pg

2-16) of PICMG_ATCA

FRONT PANEL MOUNT HOLES

MH10
Mt_Hole
MH_SIG

MH5
Mt_Hole
MH_SIG

J9

GND	2	2	1	1	GND
BP.DATA_IN_P<0>	4	4	3	3	GND
BP.DATA_IN_N<0>	6	6	5	5	GND
GND	8	8	7	7	BP.DATA_IN_P<1>
GND	10	10	9	9	BP.DATA_IN_N<1>
BP.DATA_IN_P<2>	12	12	11	11	GND
BP.DATA_IN_N<2>	14	14	13	13	GND
GND	16	16	15	15	BP.DATA_IN_P<3>
GND	18	18	17	17	BP.DATA_IN_N<3>
BP.DATA_IN_P<4>	20	20	19	19	GND
BP.DATA_IN_N<4>	22	22	21	21	GND
GND	24	24	23	23	BP.DATA_IN_P<5>
GND	26	26	25	25	BP.DATA_IN_N<5>
BP.DATA_IN_P<6>	28	28	27	27	GND
BP.DATA_IN_N<6>	30	30	29	29	GND
GND	32	32	31	31	BP.DATA_IN_P<7>
GND	34	34	33	33	BP.DATA_IN_N<7>
BP.DATA_IN_P<8>	36	36	35	35	GND
BP.DATA_IN_N<8>	38	38	37	37	GND
GND	40	40	39	39	BP.DATA_IN_P<9>
GND	42	42	41	41	BP.DATA_IN_N<9>
BP.DATA_IN_P<10>	44	44	43	43	GND
BP.DATA_IN_N<10>	46	46	45	45	GND
GND	48	48	47	47	BP.DATA_IN_P<11>
GND	50	50	49	49	BP.DATA_IN_N<11>
BP.DATA_OUT_P<0>	52	52	51	51	GND
BP.DATA_OUT_N<0>	54	54	53	53	GND
GND	56	56	55	55	BP.DATA_OUT_P<1>
GND	58	58	57	57	BP.DATA_OUT_N<1>
BP.DATA_OUT_P<2>	60	60	59	59	GND
BP.DATA_OUT_N<2>	62	62	61	61	GND
GND	64	64	63	63	BP.DATA_OUT_P<3>
GND	66	66	65	65	BP.DATA_OUT_N<3>
BP.DATA_OUT_P<4>	68	68	67	67	GND
BP.DATA_OUT_N<4>	70	70	69	69	GND
GND	72	72	71	71	BP.DATA_OUT_P<5>
GND	74	74	73	73	BP.DATA_OUT_N<5>
BP.DATA_OUT_P<6>	76	76	75	75	GND
BP.DATA_OUT_N<6>	78	78	77	77	GND
GND	80	80	79	79	BP.DATA_OUT_P<7>
GND	82	82	81	81	BP.DATA_OUT_N<7>
BP.DATA_OUT_P<8>	84	84	83	83	GND
BP.DATA_OUT_N<8>	86	86	85	85	GND
GND	88	88	87	87	BP.DATA_OUT_P<9>
GND	90	90	89	89	BP.DATA_OUT_N<9>
BP.DATA_OUT_P<10>	92	92	91	91	GND
BP.DATA_OUT_N<10>	94	94	93	93	GND
GND	96	96	95	95	BP.DATA_OUT_P<11>
GND	98	98	97	97	BP.DATA_OUT_N<11>
GND	100	100	99	99	GND
GND	102	F2	F1	101	GND

LSHM-150-03.0-L-DV-A-S-K-TR

MH8 Mt_Hole MH_SIG
MH9 Mt_Hole MH_SIG
MH6 Mt_Hole MH_SIG
MH7 Mt_Hole MH_SIG

TONGUE 2 MEZZENINE MOUNT HOLES

ISSUE	DATE	MOD NO.	DRN BY.	CHKD.	APPD.	STATUS
	Tue Oct 6 10:55:55 2026					

USED ON

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UNIVERSITY OF BRISTOL
HIGH ENERGY PHYSICS GROUP
H.H.WILLS PHYSICS LAB, TYNDALL AVE, BRISTOL, BS8 1TL.

TITLE mtca_interface_board_reocc

MODULE: 07_ backplane interface

A2

MODULE PAGE: 1 OF 1
OVERALL PAGE: 16 OF 16

TOTAL NO. OF SHEETS